



US012416300B2

(12) **United States Patent**
Moss et al.

(10) **Patent No.:** **US 12,416,300 B2**

(45) **Date of Patent:** **Sep. 16, 2025**

(54) **INFLATOR HAVING AN OUTLET
ROTATABLE RELATIVE TO A HANDLE**

USPC 173/29; 417/234
See application file for complete search history.

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(56) **References Cited**

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U.S. PATENT DOCUMENTS

2,073,014	A	3/1937	Jennings	
2,160,278	A	5/1939	William et al.	
3,099,386	A *	7/1963	Pieper	F04D 25/084 126/25 B
3,441,201	A	4/1969	Hollenberg	
3,647,323	A *	3/1972	Thomas	H02K 7/14 415/206

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(Continued)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 641 days.

FOREIGN PATENT DOCUMENTS

AU	2018200068	A1 *	7/2018		A47C 27/082
CN	106704143	A	5/2017		

(Continued)

(21) Appl. No.: **17/543,934**

(22) Filed: **Dec. 7, 2021**

OTHER PUBLICATIONS

(65) **Prior Publication Data**

US 2022/0186721 A1 Jun. 16, 2022

EP EESR dated, Jun. 22, 2022 in corresponding EP application No.
21214191.5.

Related U.S. Application Data

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(60) Provisional application No. 63/126,133, filed on Dec.
16, 2020.

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(51) **Int. Cl.**

F04B 39/12 (2006.01)

B25F 3/00 (2006.01)

B25F 5/02 (2006.01)

(57)

ABSTRACT

(52) **U.S. Cl.**

CPC **F04B 39/12** (2013.01); **B25F 5/02**
(2013.01); **B25F 3/00** (2013.01)

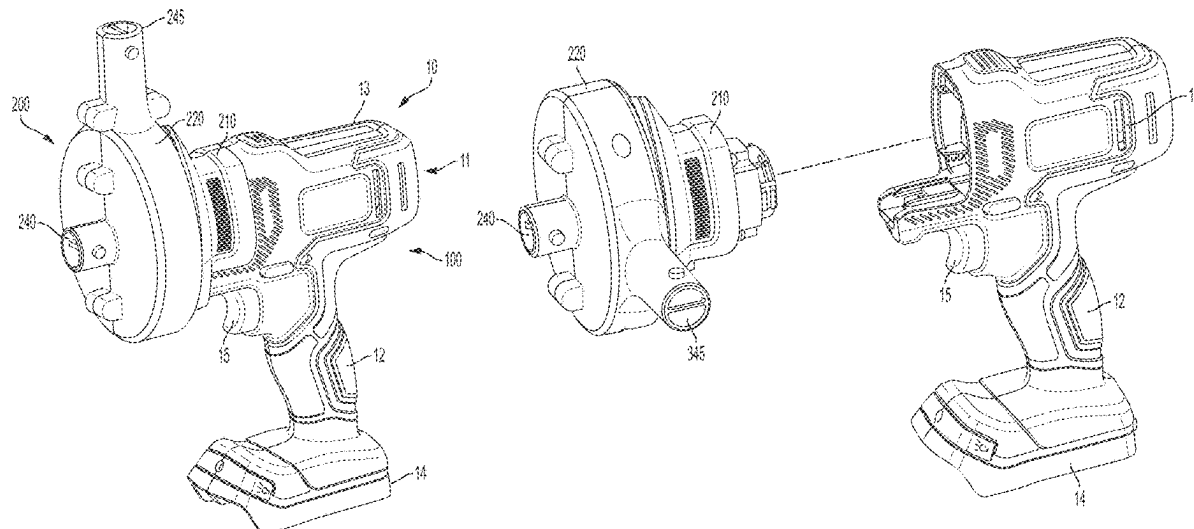
An inflator includes a housing, a handle formed by the
housing, a motor housed in the housing, a trigger configured
to activate the motor, an inlet having an inlet axis that is
generally parallel or concentric with a longitudinal axis of
the motor, an outlet which is transverse to the inlet and a fan
configured to draw air in through the inlet and blow air out
through the outlet. The inlet is generally transverse to the
handle.

The position of the outlet is rotatable relative to the handle.

(58) **Field of Classification Search**

CPC B25F 5/027; B25F 5/02; B25F 3/00; F05D
2230/52; F04B 39/12; F04B 39/14; F04B
35/04; F04D 25/0673; F04D 25/084;
F04D 29/62

11 Claims, 18 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

3,809,504	A *	5/1974	Goossens		F04D 25/084	415/206
4,021,150	A *	5/1977	Mabuchi		F04D 13/068	415/121.2
4,287,795	A *	9/1981	Curtiss		B25B 21/002	81/57.3
4,413,371	A *	11/1983	Tuggle		A47L 5/14	15/328
4,466,780	A	8/1984	Naurath			
4,516,561	A *	5/1985	Stawski		F04D 25/08	126/25 B
4,678,014	A	7/1987	Owen et al.			
4,884,314	A *	12/1989	Miner		A01G 20/47	15/419
D391,969	S	3/1998	Poon			
5,848,875	A	12/1998	San-Jou			
5,879,141	A	3/1999	Yokozawa et al.			
5,890,882	A *	4/1999	Feldman		F04D 25/0673	5/713
5,938,410	A *	8/1999	Lee		A47L 5/24	417/423.2
D415,500	S	10/1999	Poon			
D422,602	S	4/2000	Poon			
6,045,341	A *	4/2000	Mak		A63H 33/40	446/179
6,119,738	A *	9/2000	Idol		F04D 25/084	239/525
6,155,781	A	12/2000	Tsai			
6,290,469	B1	9/2001	Archibald			
6,295,693	B1 *	10/2001	Chou		F04D 29/626	141/66
D451,521	S	12/2001	Poon			
6,468,047	B1	10/2002	Huang et al.			
D466,908	S	12/2002	Poon			
D467,941	S	12/2002	Duncan			
D467,942	S	12/2002	Duncan			
D470,863	S	2/2003	Williams			
D486,159	S	2/2004	Poon			
7,210,902	B2 *	5/2007	Song		F04D 29/503	415/206
8,210,834	B2	7/2012	Tsai			
8,613,605	B2	12/2013	Hammar et al.			
8,672,650	B2	3/2014	Horng et al.			
8,702,403	B2	4/2014	Horng et al.			
8,721,298	B2 *	5/2014	Tanimoto		F04D 25/0653	417/234
8,974,194	B2	3/2015	Yin et al.			
8,974,195	B2	3/2015	Horng et al.			
9,239,060	B2	1/2016	Heymann			
9,291,170	B2	3/2016	Heymann			
9,371,828	B2 *	6/2016	Tsai		F04D 27/008	
9,388,826	B2	7/2016	Liu et al.			
D771,713	S	11/2016	Ye et al.			
9,655,278	B2	5/2017	Wu			
9,702,367	B2	7/2017	Wu			
9,707,580	B1 *	7/2017	Mancl		H02P 25/14	
9,883,634	B2 *	2/2018	Romito		A01G 20/47	
D817,364	S	5/2018	He			
D840,437	S	2/2019	Blaser			
D841,696	S	2/2019	Blaser			
D868,119	S	11/2019	Kobler et al.			
D869,507	S	12/2019	Huang et al.			
D869,508	S	12/2019	Huang et al.			
2005/0042051	A1	2/2005	Obermeier et al.			
2006/0073040	A1	4/2006	Metzger et al.			
2008/0226476	A1 *	9/2008	Lau		F04D 25/084	417/411
2008/0286124	A1	11/2008	Sen et al.			
2010/0170538	A1 *	7/2010	Baker		B08B 5/04	409/137
2013/0020106	A1	1/2013	Kuehne et al.			
2014/0064943	A1 *	3/2014	Wang		F04B 39/121	415/182.1
2014/0332243	A1 *	11/2014	Baskar		B25F 5/02	173/29
2018/0187687	A1 *	7/2018	Yakubova		F04D 25/08	
2019/0105458	A1 *	4/2019	Hammes		A61L 9/122	
2019/0290080	A1 *	9/2019	Liu		B08B 5/02	
2021/0340991	A1 *	11/2021	Ojima		F04D 25/0606	

FOREIGN PATENT DOCUMENTS

CN	207268209	U	4/2018	
CN	210317647	U	4/2020	
CN	211009180	U	7/2020	
DE	20100015	U1 *	6/2001	 F04B 35/04
DE	202012100512	U1	3/2012	
DE	102012211038	A1	1/2014	
EP	3346126	A1	7/2018	
IN	207223899	U	4/2018	
NO	2014161257	A1	10/2014	

* cited by examiner

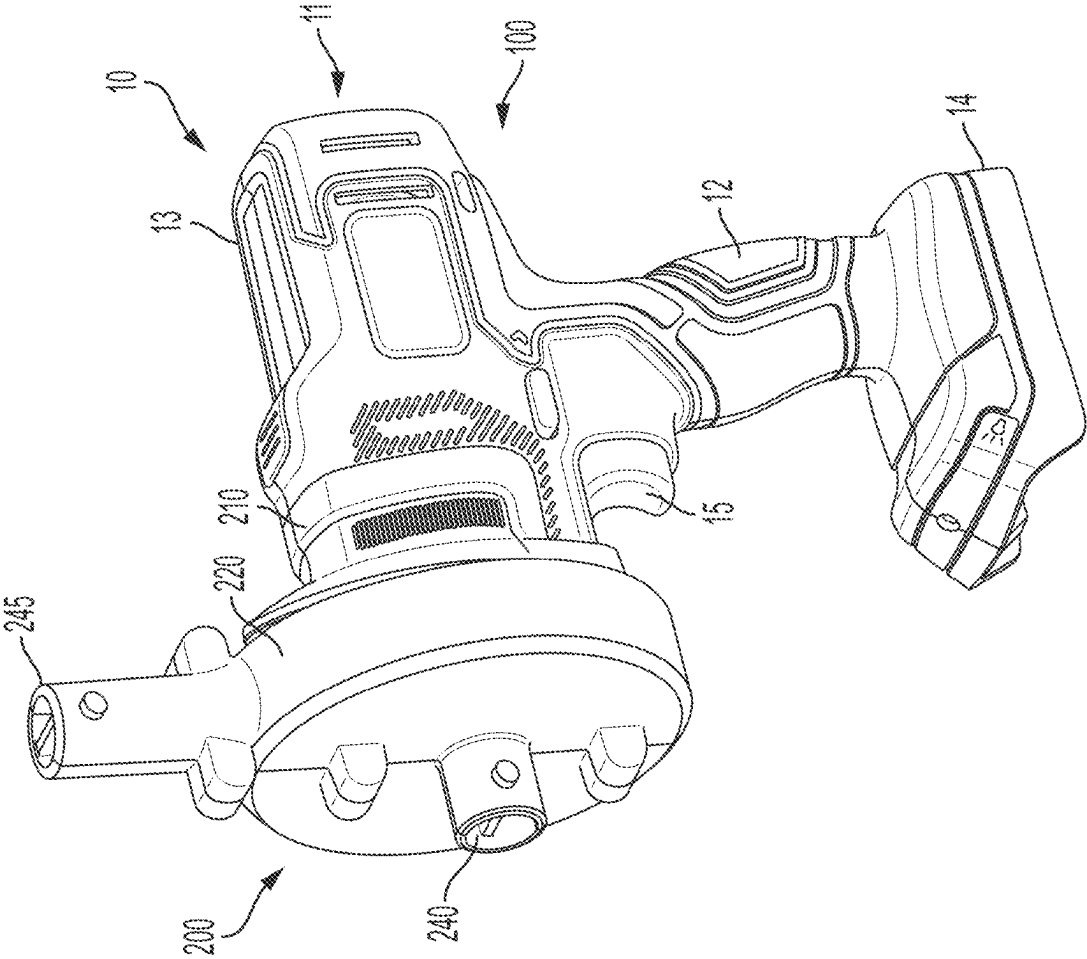


FIG. 1

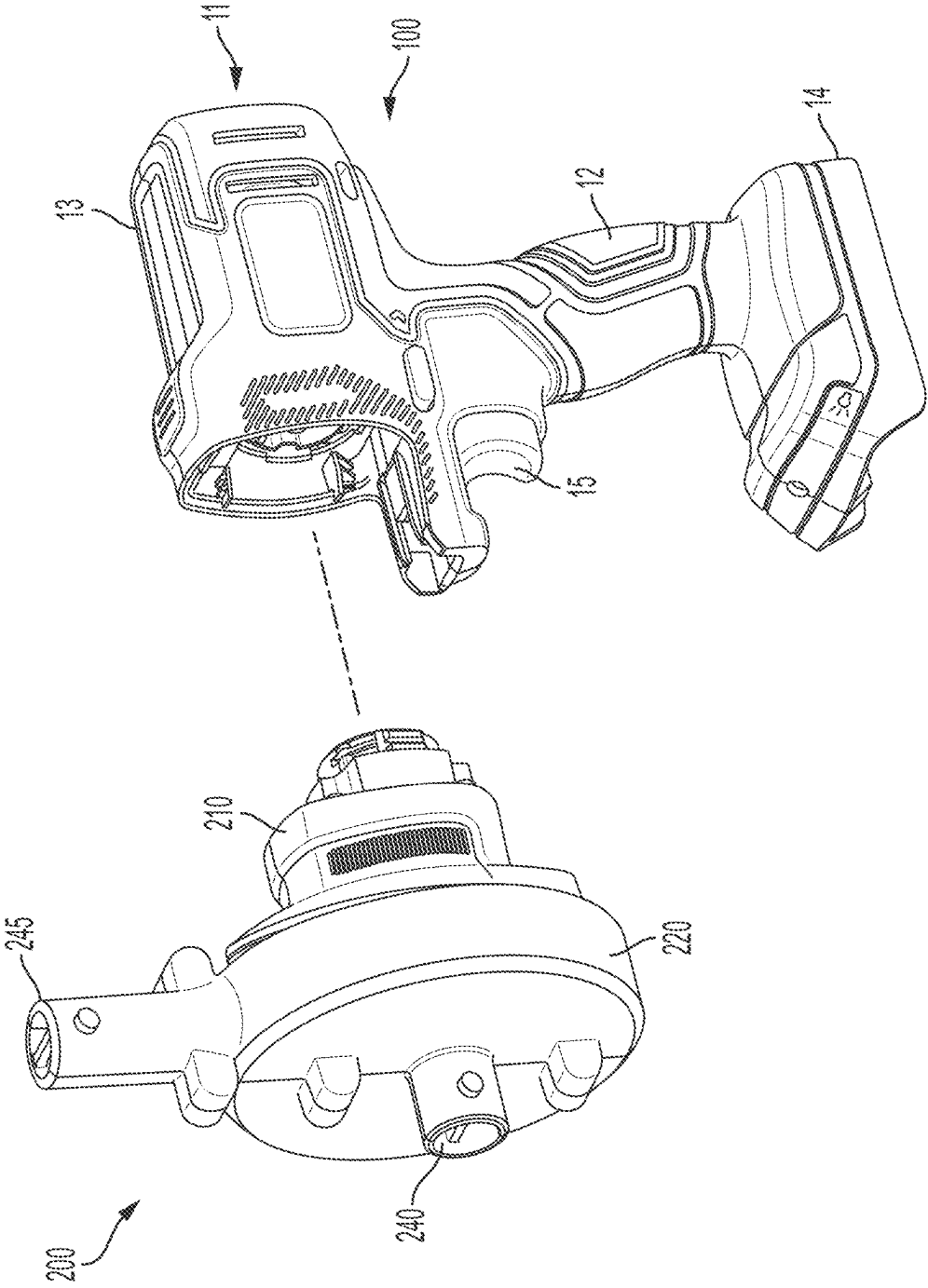


FIG. 2

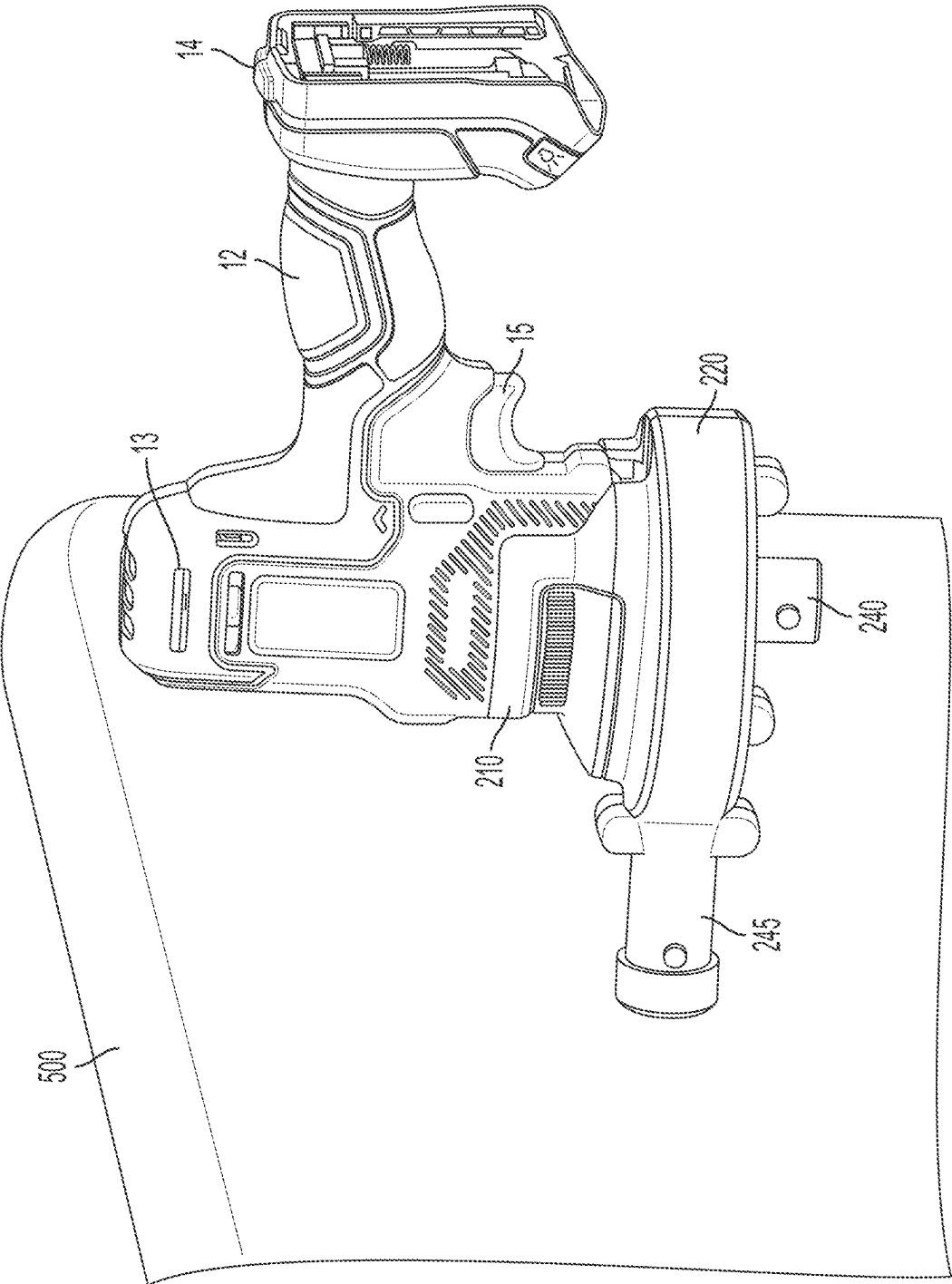


FIG. 3

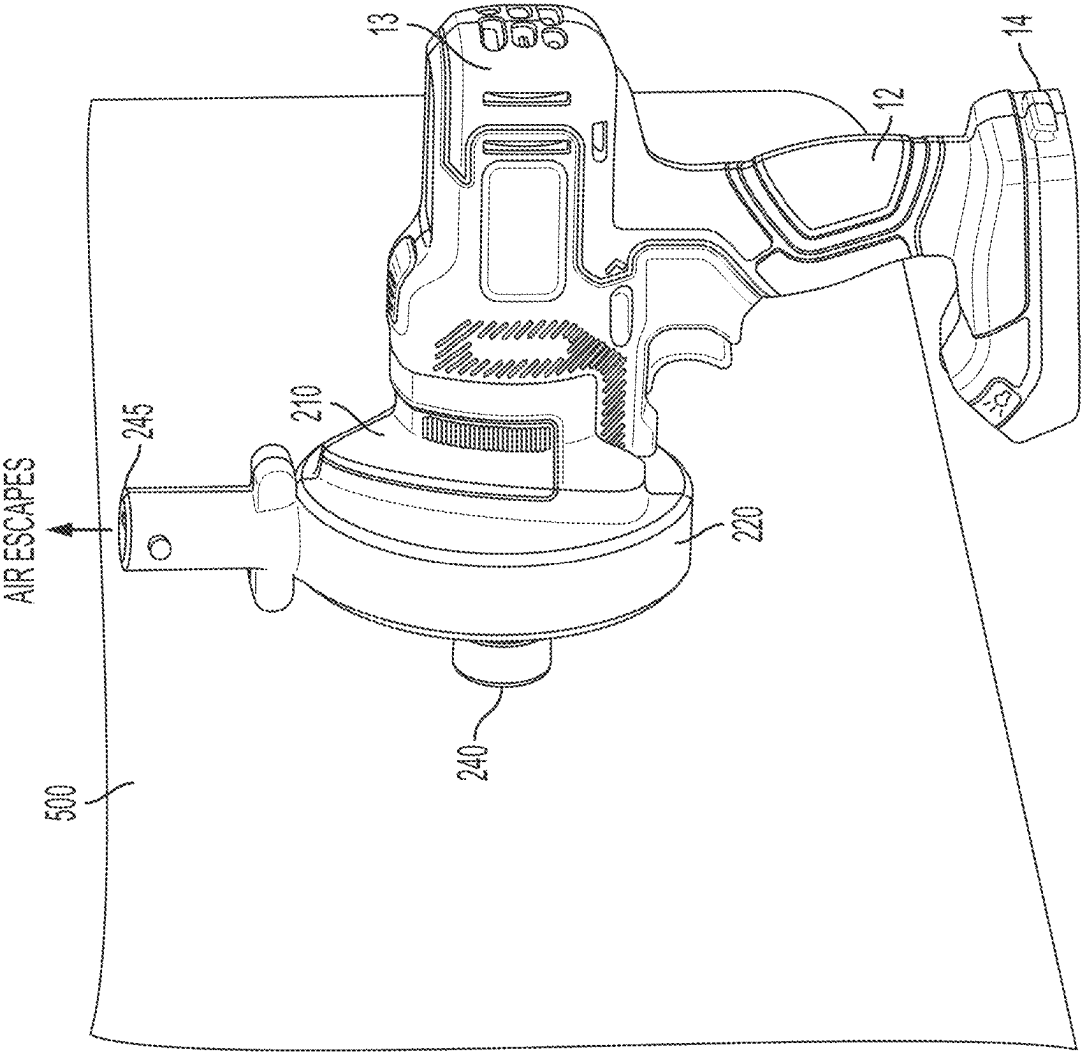


FIG. 4

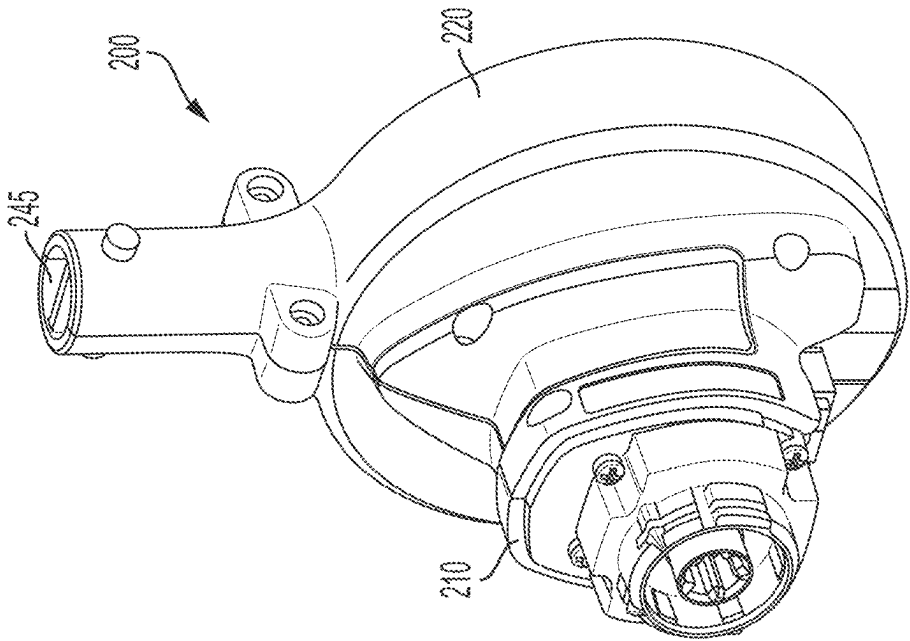


FIG. 6

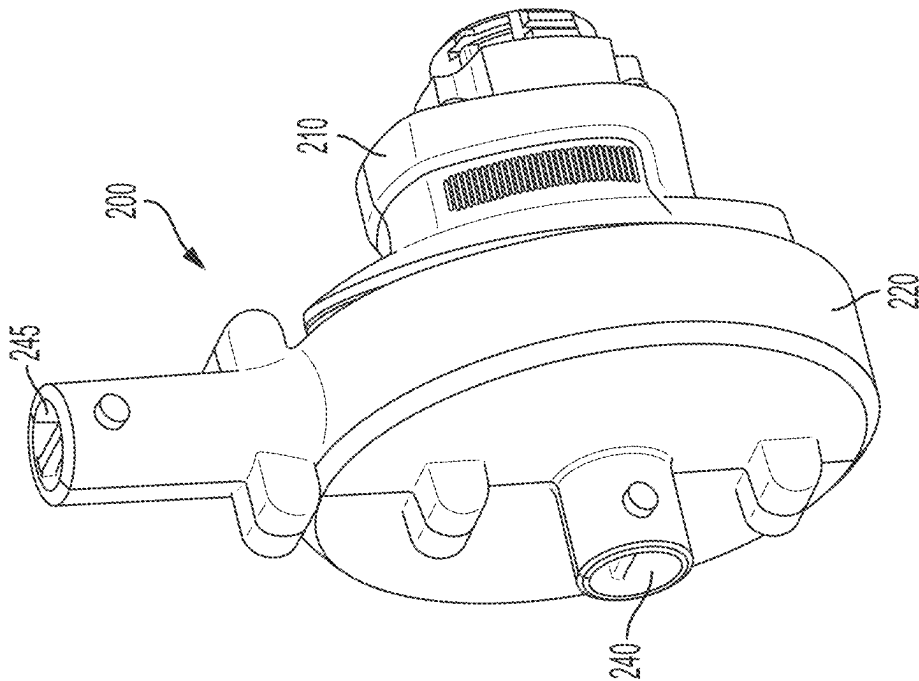


FIG. 5

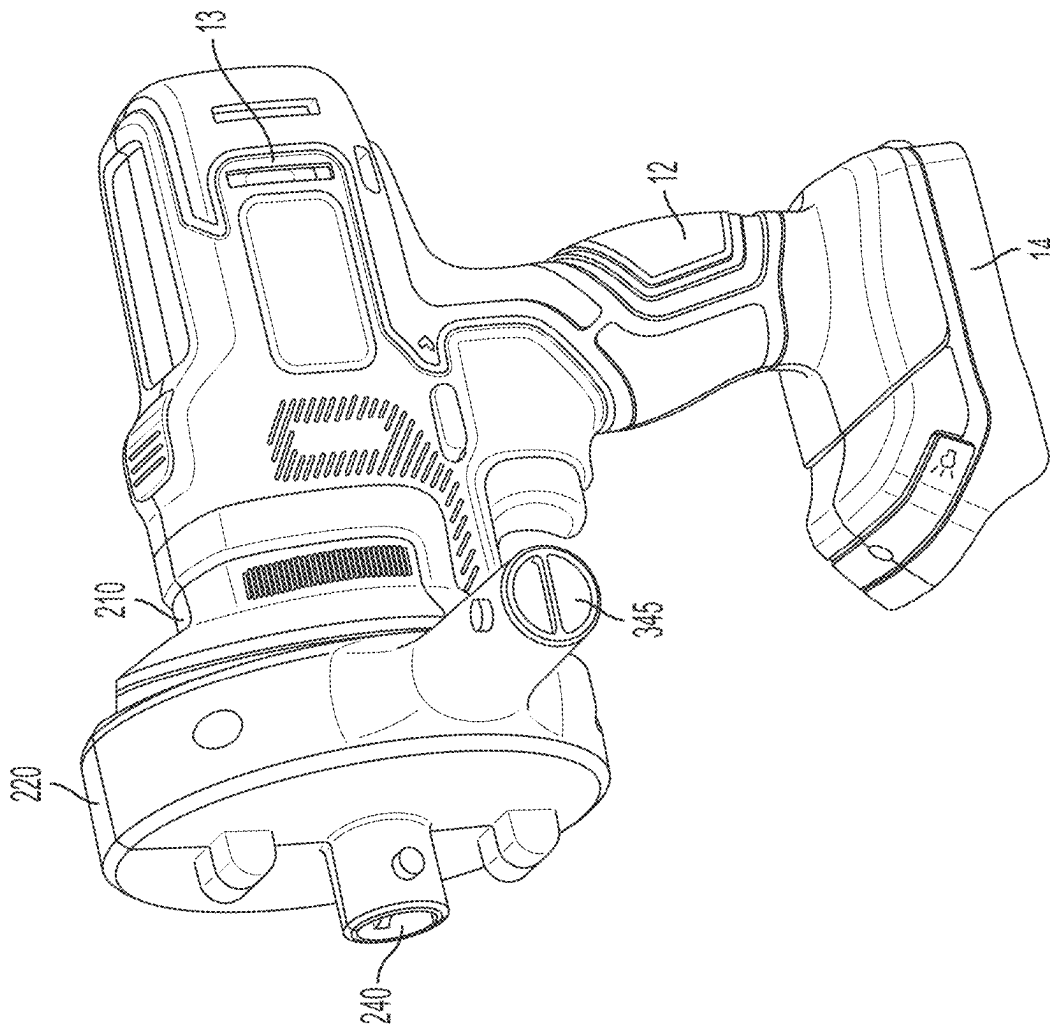


FIG. 7

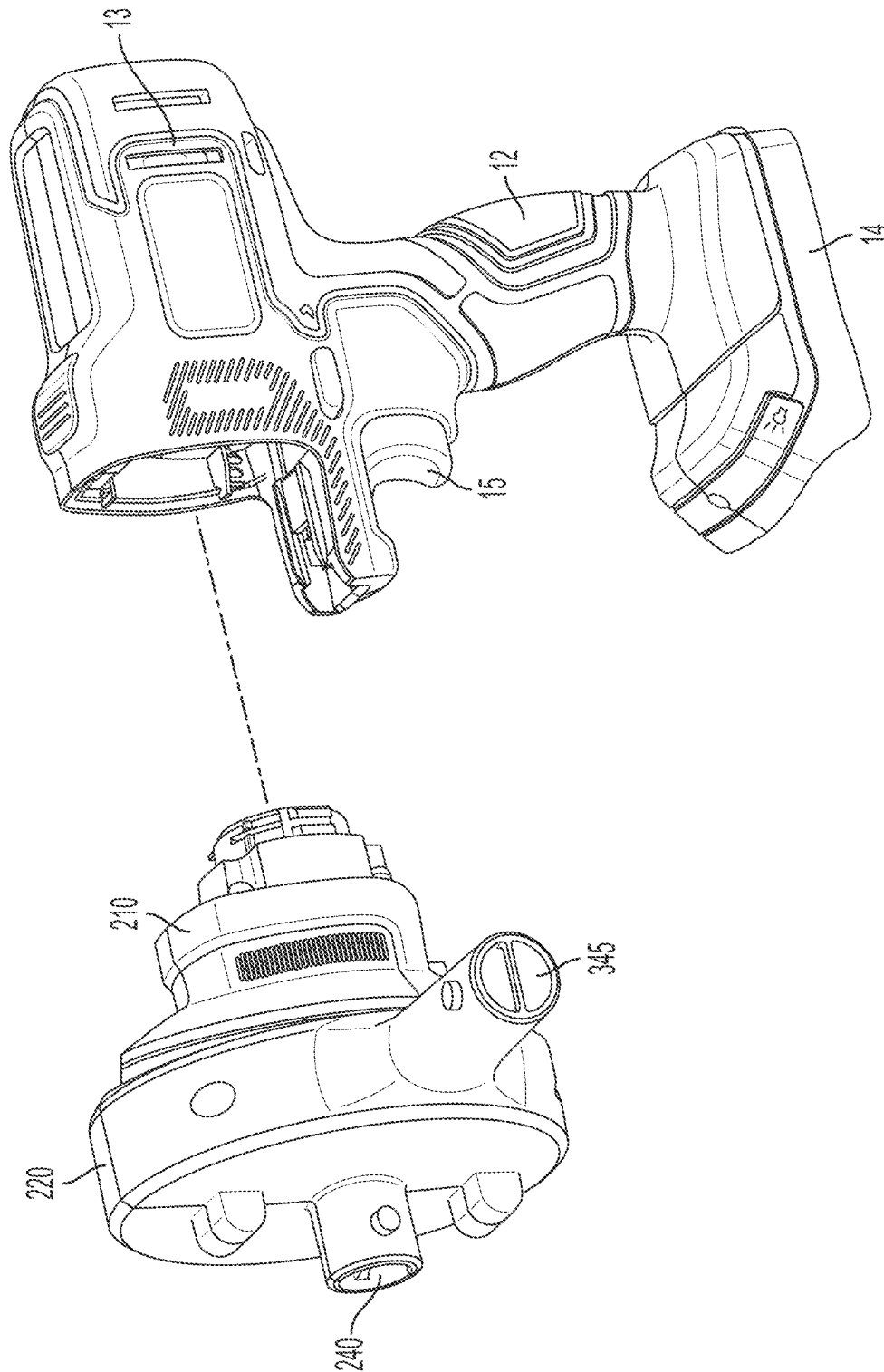


FIG. 8

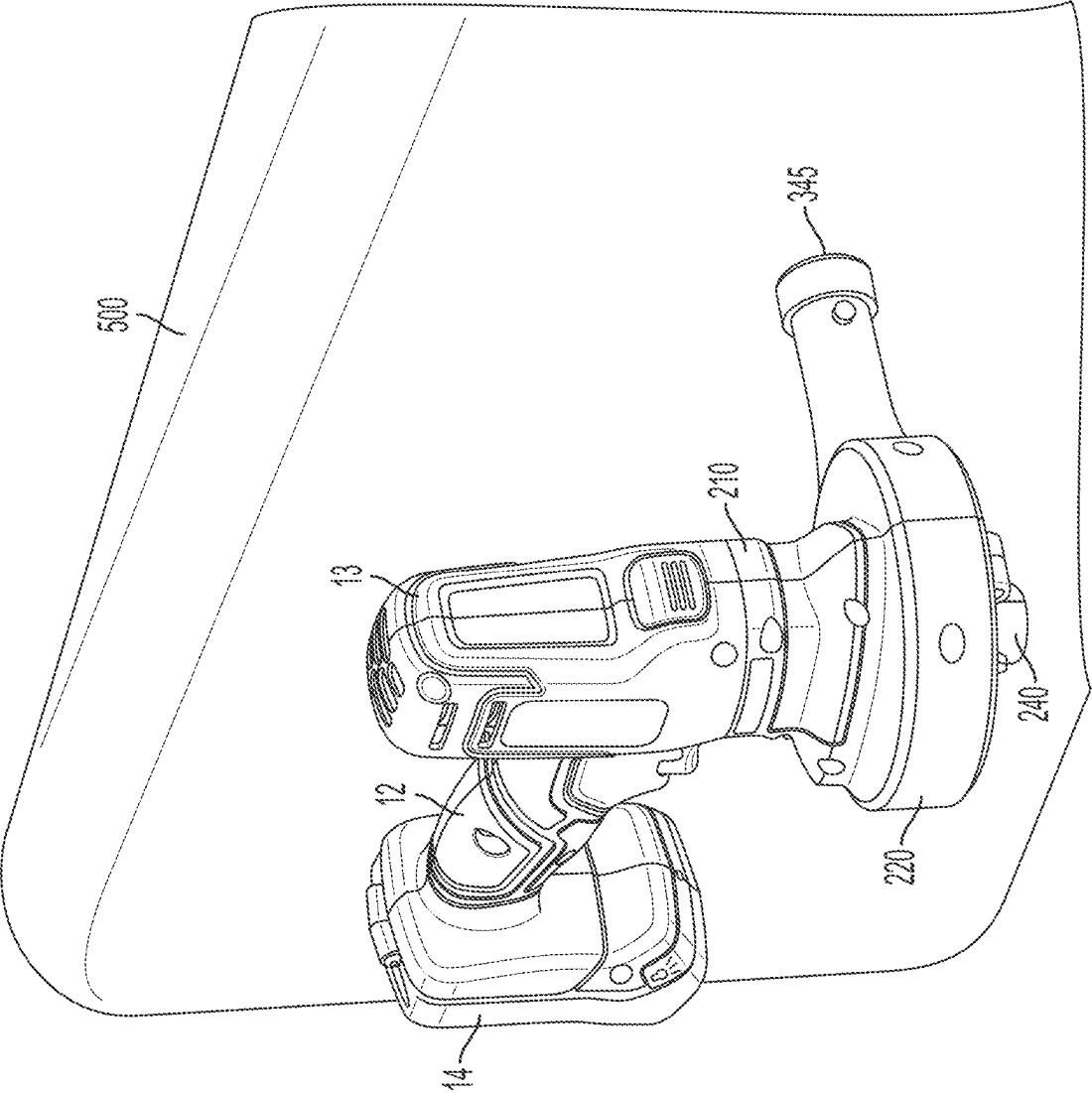


FIG. 9

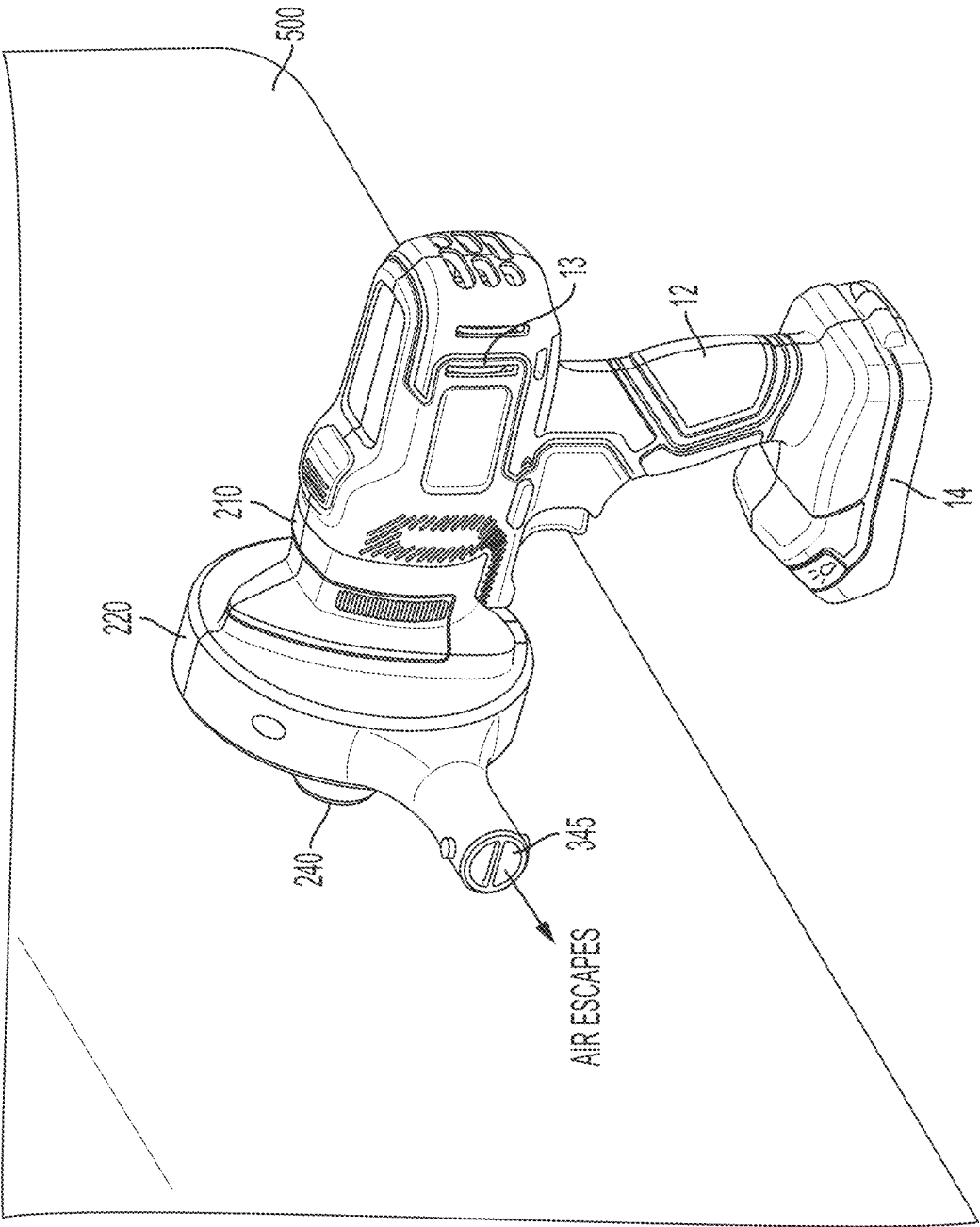


FIG. 10

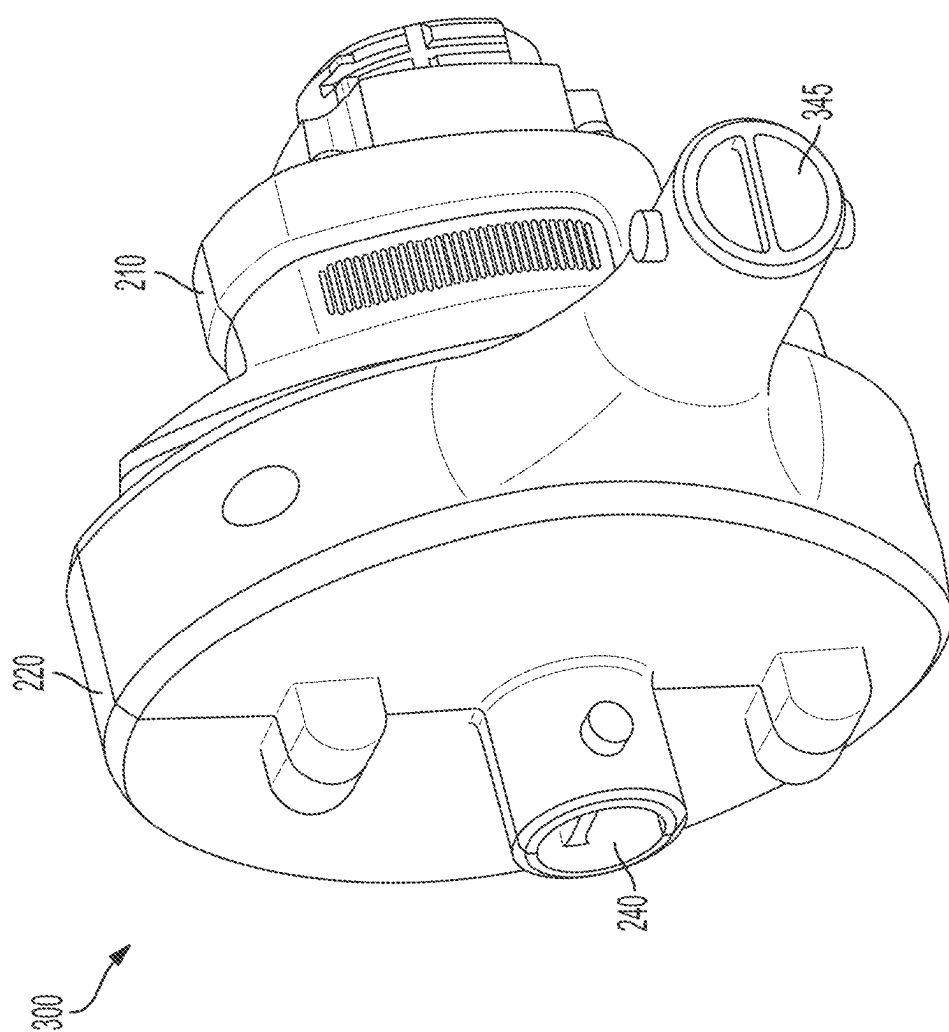


FIG. 11

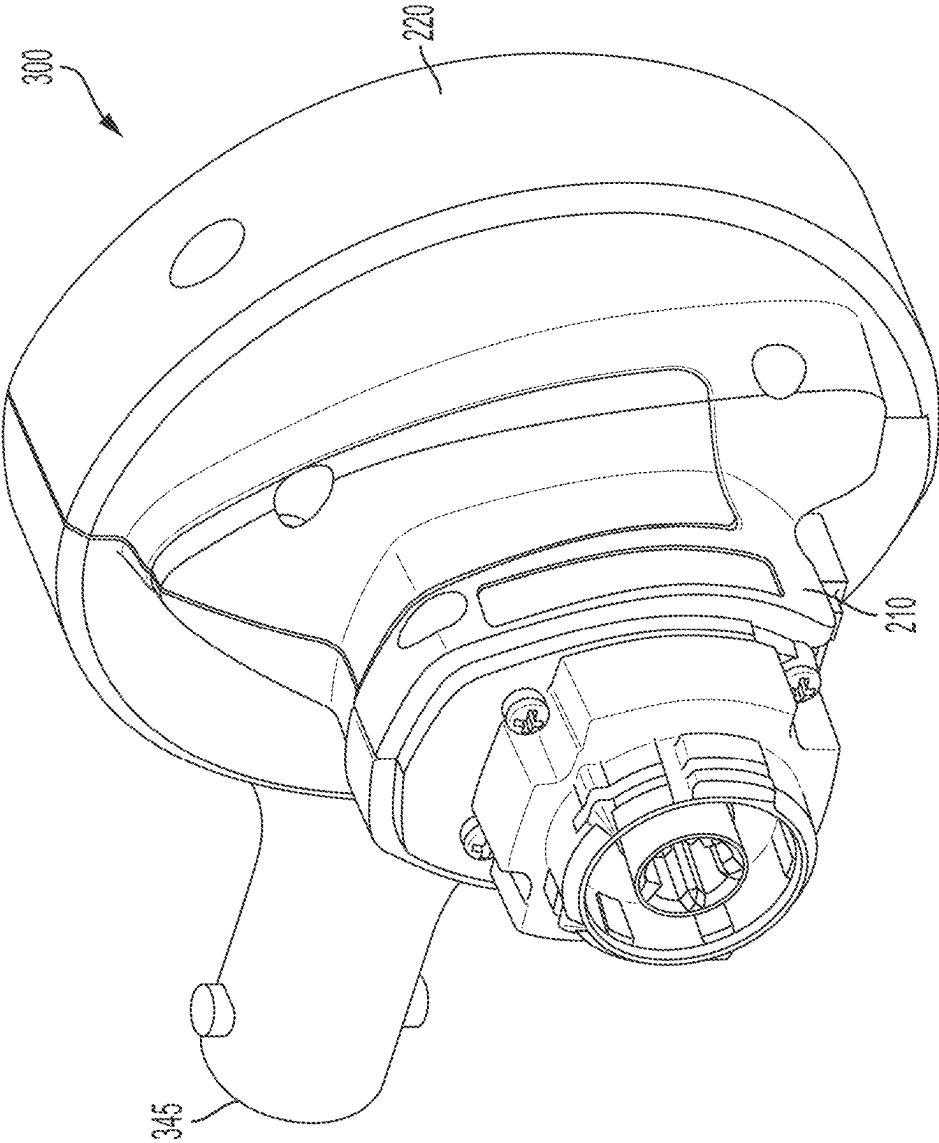


FIG. 12

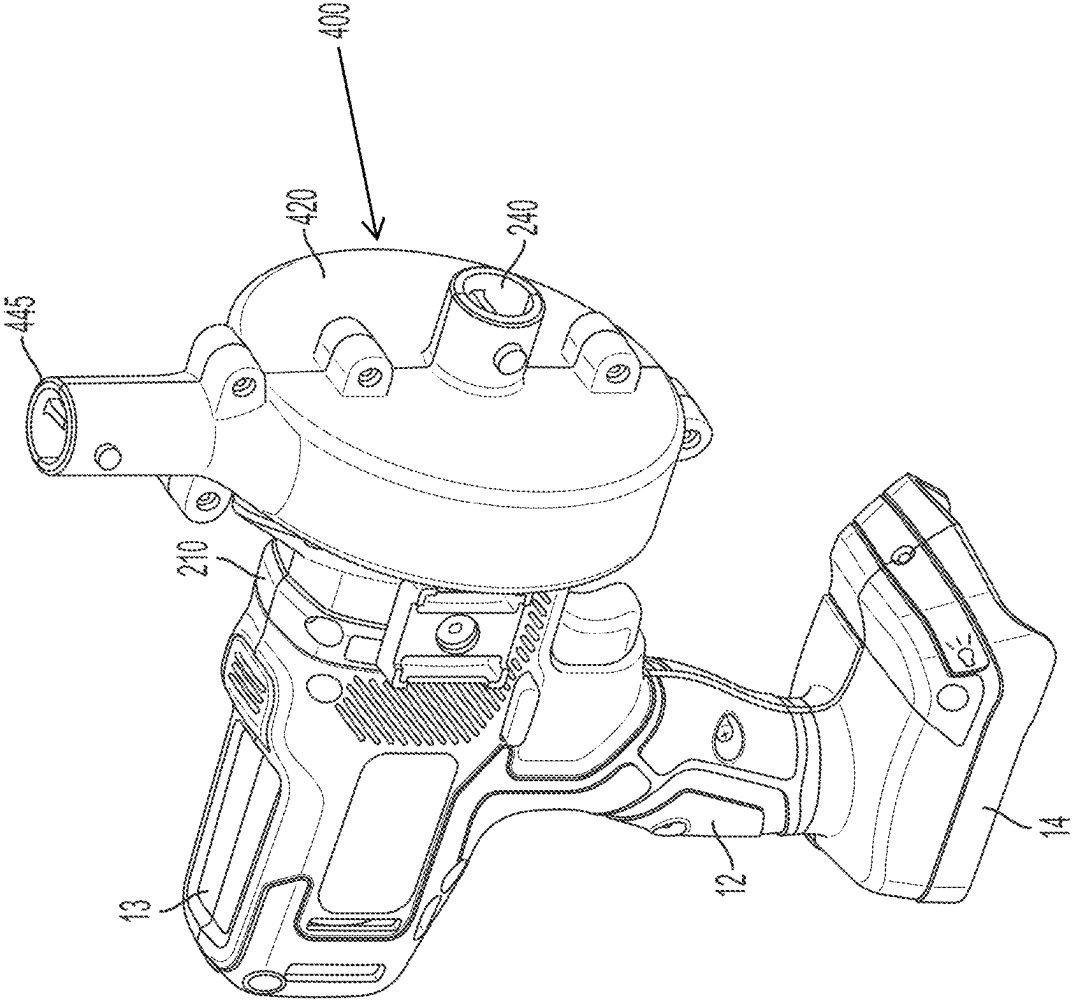


FIG. 13

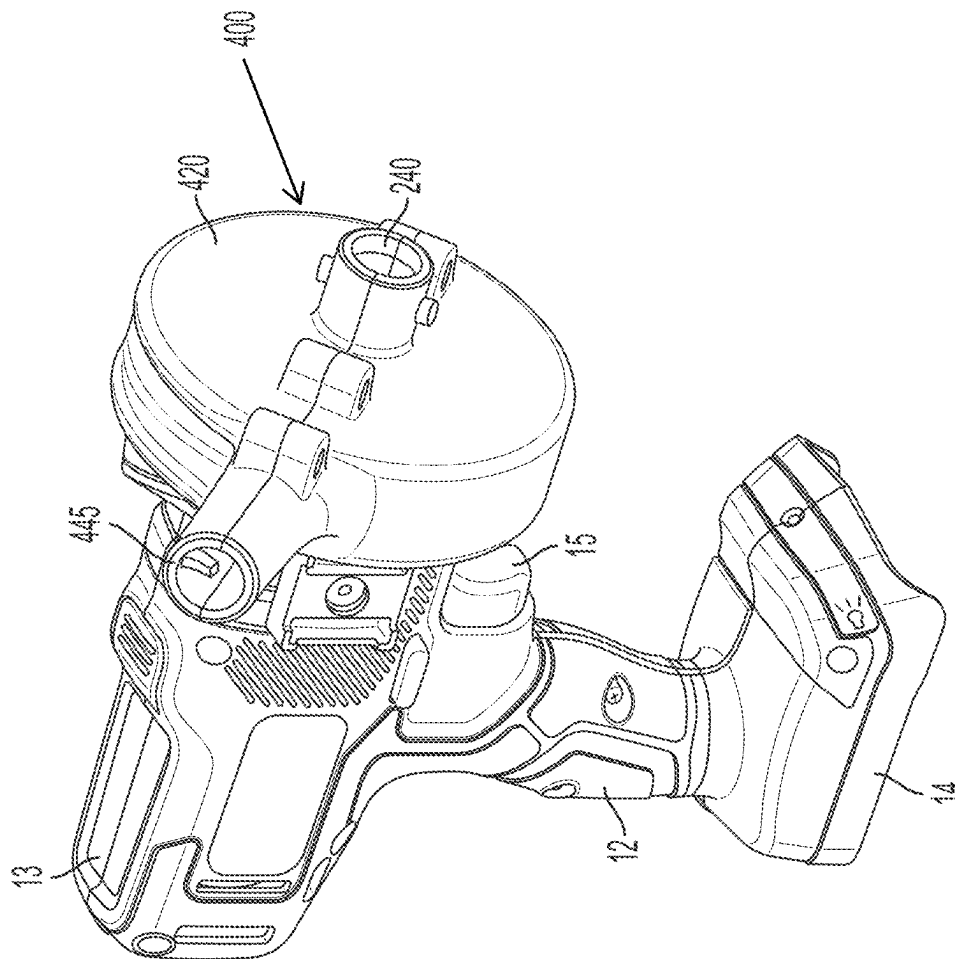


FIG. 14

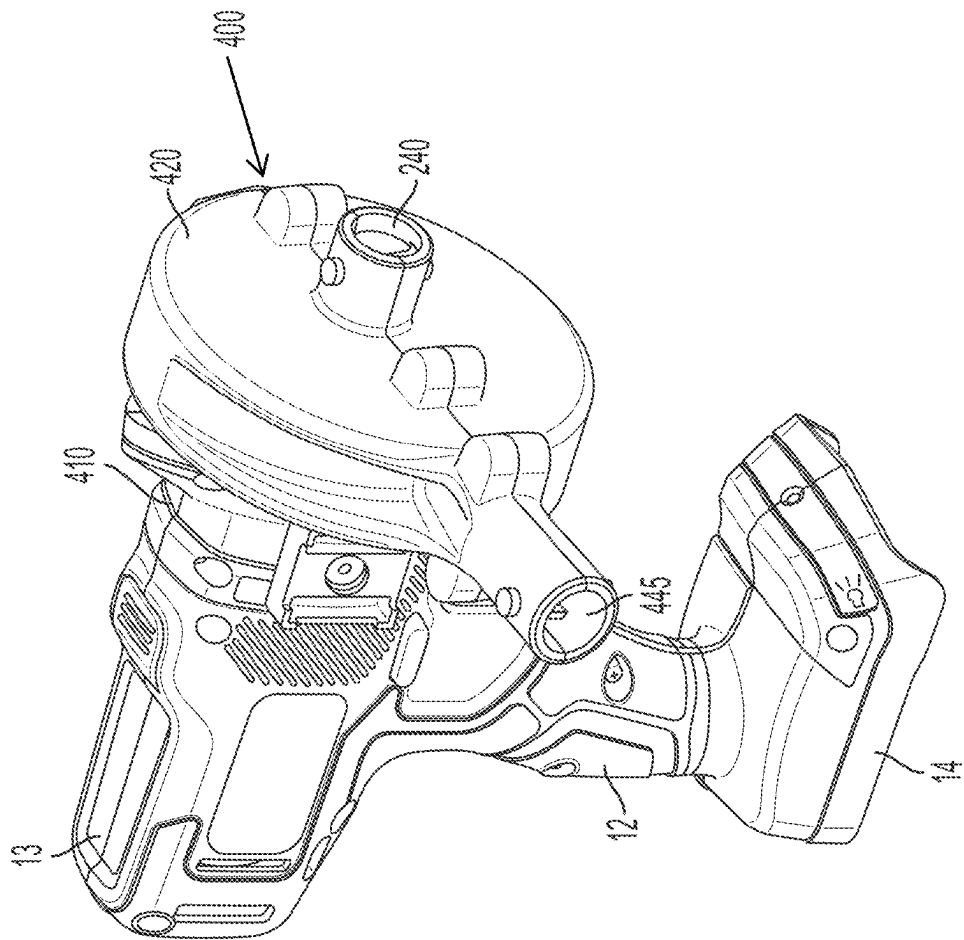


FIG. 15

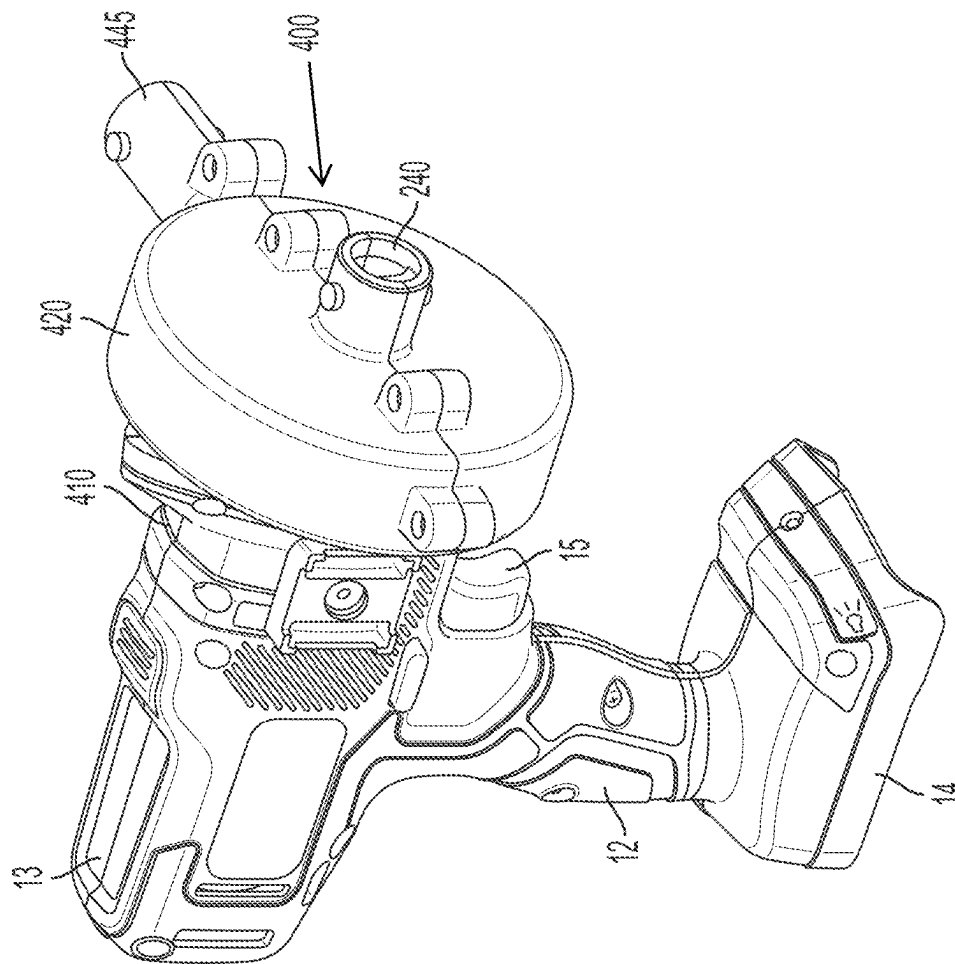


FIG. 16

REPLACEMENT SHEET
16/18

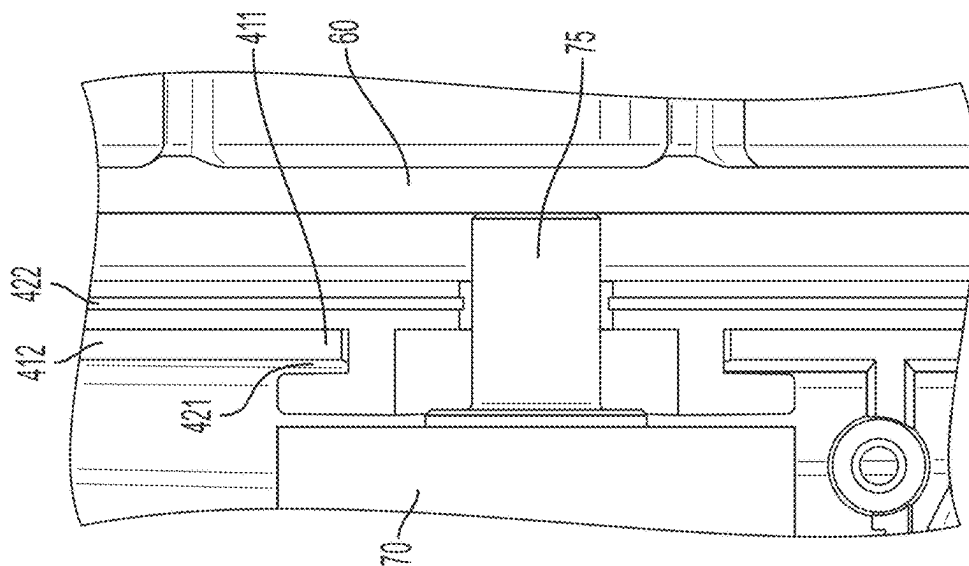


FIG. 18

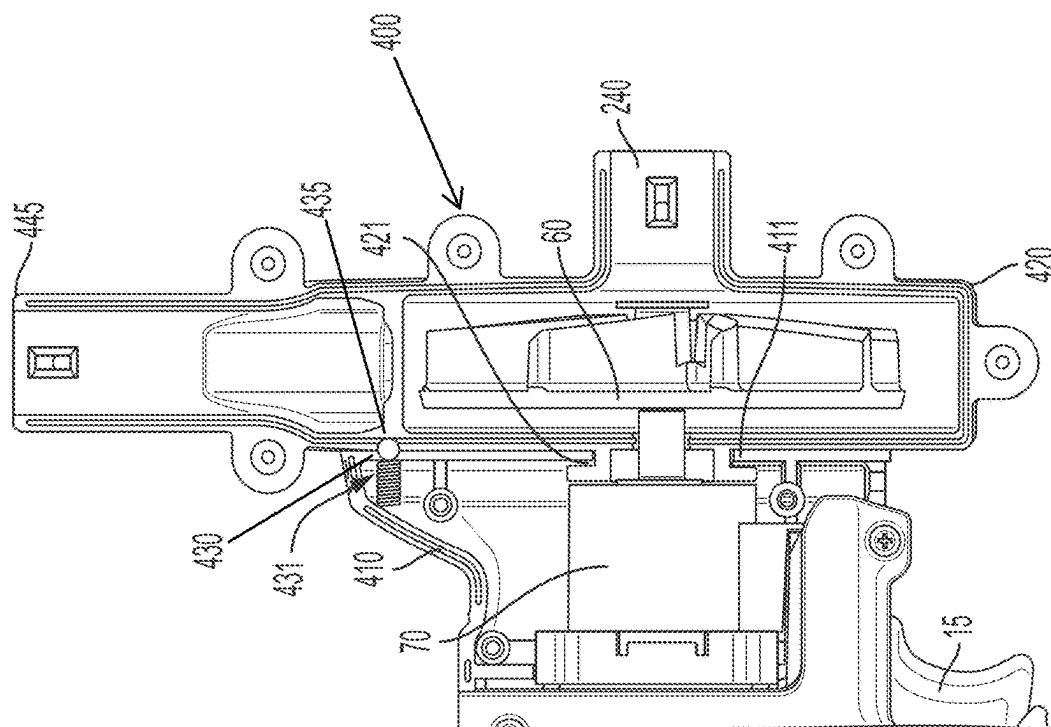
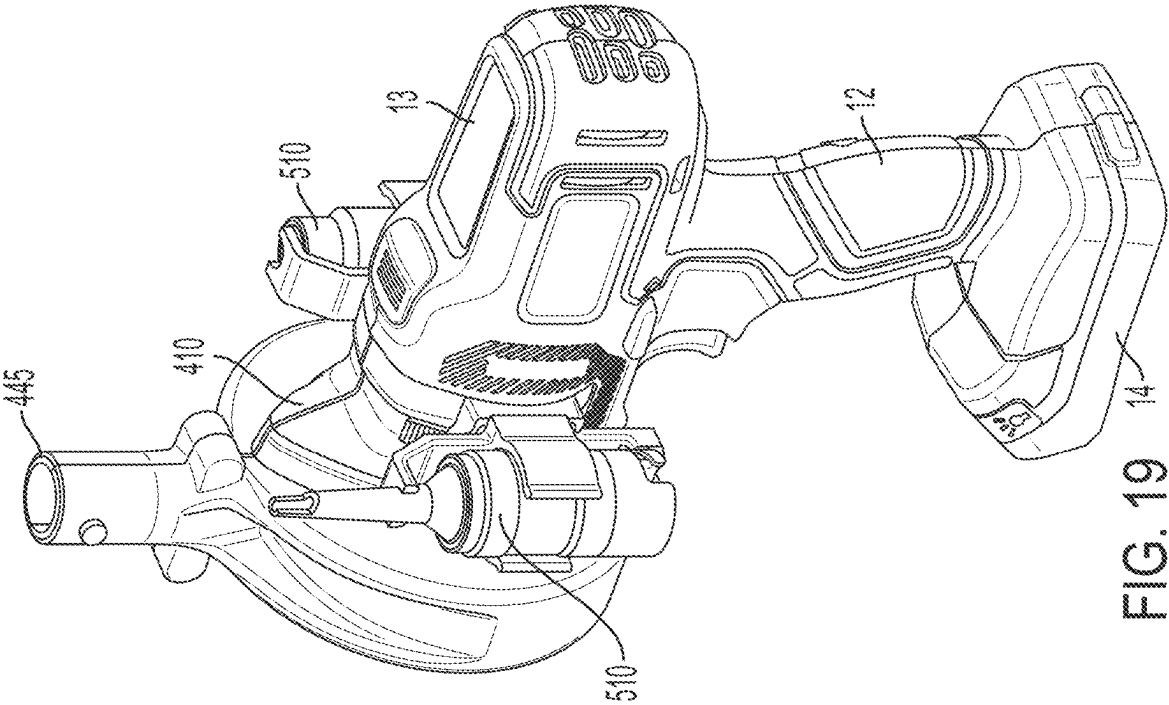


FIG. 17



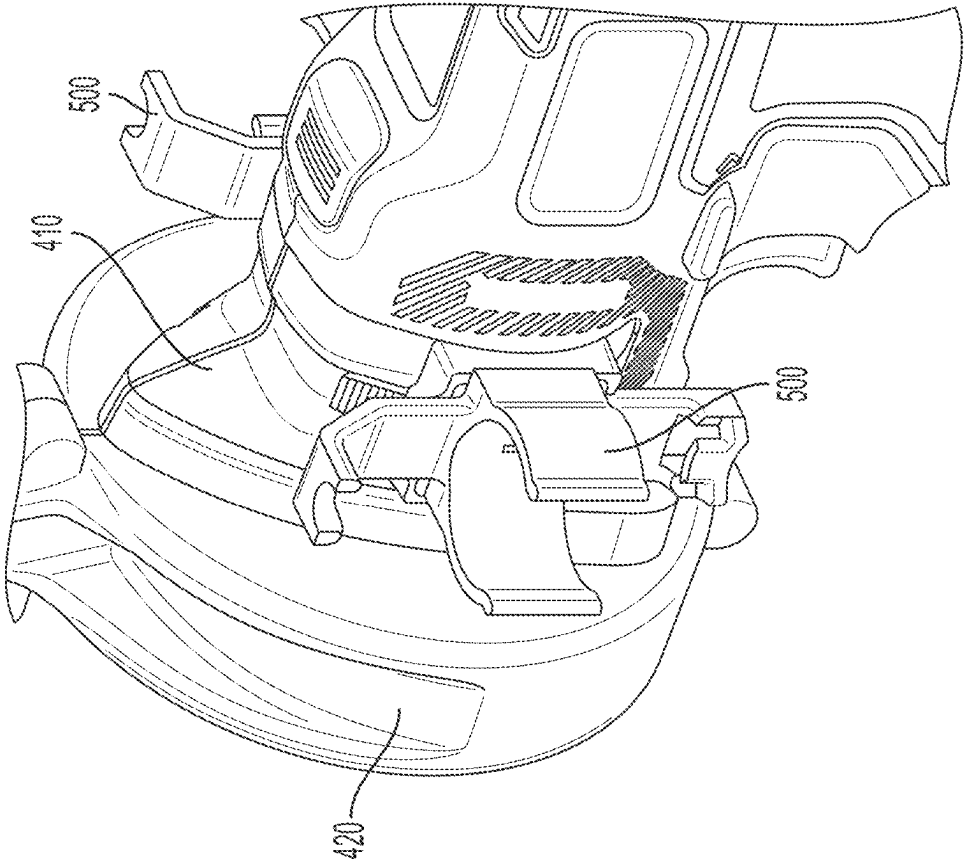


FIG. 20

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INFLATOR HAVING AN OUTLET ROTATABLE RELATIVE TO A HANDLE

This application claims the benefit of U.S. Provisional Application No. 63/126,133 filed on Dec. 16, 2020, entitled High Volume, Low Pressure Inflator. The entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

The present disclosure relates to an inflator. In particular, exemplary embodiments disclose a high flow, low pressure inflator.

SUMMARY OF EMBODIMENTS OF THE INVENTION

Aspects of the present disclosure relate to an exemplary embodiment of an inflator power tool.

According to one aspect, there is an exemplary embodiment of a high-volume, low pressure inflator that includes a housing; a handle formed by the housing; a motor housed in the housing; a trigger configured to activate the motor; an inlet having an inlet axis that is generally parallel or concentric with a longitudinal axis of the motor; an outlet which is transverse to the inlet; a fan configured to draw air in through the inlet and blow air out through the outlet.

The inlet may be generally transverse to the handle.

The position of the outlet may be rotatable relative to the handle.

The outlet may be rotatable about a rotation axis.

The rotation axis may be generally concentric or parallel to the longitudinal axis of the motor.

The rotation axis may be generally transverse to the handle.

According to another aspect, there is an exemplary embodiment of an inflator, the inflator including a tool base unit, the tool base unit comprising a handle, a motor and a trigger; and an inflator tool head removably attached to the tool base unit.

The inflator tool head may include an air inlet, an air outlet and a fan configured to draw air in through the air inlet and blow air out through the air outlet.

The inlet may be transverse to the outlet.

The outlet may be rotatable with respect to the tool base unit while the inflator tool head is attached to the tool base unit.

The inlet may be generally transverse to the handle.

The inflator may further include an attachment holder configured to hold an inflator attachment.

The attachment holder may be disposed on the inflator tool head.

The inflator tool head may include a connection portion and a fan housing portion.

The fan may be disposed in the fan housing portion.

The fan housing portion may be rotatable with respect to the connection portion.

The fan housing portion may include the air outlet.

The fan housing portion may include the air inlet.

The fan housing portion may be configured to be held in place in a variety of rotational positions with respect to the connection portion.

The fan housing portion may be configured to be held in place by friction.

The fan housing portion may be configured to be held in place by a biasing assembly.

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The fan housing portion may be configured to be held in place at plurality of distinct rotational positions.

The plurality of distinct rotational positions may include at least four distinct rotational positions.

The connection portion may house a connector which receives drive from the motor.

According to another aspect, there is an exemplary embodiment of a high-volume, low pressure inflator, the inflator including a housing; a handle formed by the housing; a motor housed in the housing; a trigger configured to activate the motor; an inflator head portion; an inlet having an inlet axis that is generally parallel or concentric with a longitudinal axis of the motor; an outlet which is transverse to the inlet; a fan configured to draw air in through the inlet and blow air out through the outlet.

The inlet may be generally transverse to the handle.

The position of the outlet may be rotatable relative to the handle.

The inflator head portion may include a connection portion and a fan housing portion.

The fan may be disposed in the fan housing portion.

The fan housing portion may include the outlet and is rotatable with respect to the connection portion.

The outlet may be rotatable about a rotation axis.

The rotation axis may be generally concentric or parallel to the longitudinal axis of the motor.

The rotation axis may be generally transverse to the handle.

These and other aspects of various embodiments of the present invention, as well as the methods of operation and functions of the related elements of structure and the combination of parts and economies of manufacture, will become more apparent upon consideration of the following description and the appended claims with reference to the accompanying drawings, all of which form a part of this specification, wherein like reference numerals designate corresponding parts in the various figures. In one embodiment of the invention, the structural components illustrated herein are drawn to scale. It is to be expressly understood, however, that the drawings are for the purpose of illustration and description only and are not intended as a definition of the limits of the invention. In addition, it should be appreciated that structural features shown or described in any one embodiment herein can be used in other embodiments as well. As used in the specification and in the claims, the singular form of "a", "an", and "the" include plural referents unless the context clearly dictates otherwise.

All closed-ended (e.g., between A and B) and open-ended (greater than C) ranges of values disclosed herein explicitly include all ranges that fall within or nest within such ranges. For example, a disclosed range of 1-10 is understood as also disclosing, among other ranged, 2-10, 1-9, 3-9, etc.

BRIEF DESCRIPTION OF THE DRAWINGS

For a better understanding of embodiments of the present invention as well as other objects and further features thereof, reference is made to the following description which is to be used in conjunction with the accompanying drawings, where:

FIG. 1 is a perspective view of an inflator according to a first exemplary embodiment of the present application;

FIG. 2 is a perspective view of the exemplary embodiment of the inflator with the inflator tool head removed from the tool base unit;

FIG. 3 is a perspective view of the inflator inflating a mattress;

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FIG. 4 is a perspective view of the inflator deflating the mattress;

FIG. 5 is a perspective view the inflator tool head of the exemplary embodiment of the inflator;

FIG. 6 is another perspective view the inflator tool head of the exemplary embodiment of the inflator;

FIG. 7 is a perspective view of an inflator according to a second exemplary embodiment of the present application;

FIG. 8 is a perspective view of the second exemplary embodiment of the inflator with the inflator tool head removed from the tool base unit;

FIG. 9 is a perspective view of the second exemplary embodiment of the inflator inflating a mattress;

FIG. 10 is a perspective view of the inflator deflating the mattress;

FIG. 11 is a perspective view the inflator tool head of the exemplary embodiment of the inflator;

FIG. 12 is another perspective view the inflator tool head of the exemplary embodiment of the inflator;

FIG. 13 is a perspective view of an inflator according to a third exemplary embodiment;

FIG. 14 is another perspective view of the inflator according to the third exemplary embodiment;

FIG. 15 is another perspective view of the inflator according to the third exemplary embodiment;

FIG. 16 is another perspective view of the inflator according to the third exemplary embodiment;

FIG. 17 is a close-up cut-away view of the inflator according to the third exemplary embodiment;

FIG. 18 is another close-up cut-away view of an inflator according to the third exemplary embodiment;

FIG. 19 is a perspective view of the inflator with attachment holders; and

FIG. 20 is a close-up perspective view of the inflator with attachment holders.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS OF THE INVENTION

FIG. 1 illustrates a first exemplary embodiment of inflator 10. The inflator 10 of the exemplary embodiment is a high volume, low pressure inflator of that provides a high volume of airflow for inflating air mattresses, rafts and other inflatable items. As shown in FIG. 1, the inflator 10 includes a housing 11. The housing 11 forms a handle 12 and a motor housing portion 13. A motor is housed in the motor housing portion 13. There may be a transmission connected to the motor. The housing 11 also has a battery receiving portion 14 configured to receive a battery pack. In this case, the battery pack may be a power tool battery pack configured to power a variety of power tools such as drills, sanders and saws. The battery pack may be of the type shown in, for example, U.S. Pat. Nos. 7,598,705; 7,661,486; or U.S. Patent Application Publication No. 2018/0331335. U.S. Pat. Nos. 7,598,705; 7,661,486; and U.S. Patent Application Publication No. 2018/0331335 are hereby incorporated by reference. The battery pack powers the inflator 10 and particularly the motor of the inflator 10. A user-actuable trigger 15 is housed in the handle 12. The trigger 15 can be actuated to actuate the motor.

The inflator 10 of the first exemplary embodiment includes a tool base unit 100 and an inflator tool head 200. As is shown in FIG. 2 of the present application, the inflator tool head 200 is removably attached to the tool base unit 100. Other tool heads such as a drill tool head, saw tool head, sander tool head and others may also be attached to the tool base unit 100 forming a multi-headed tool system. The tool

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base unit 100 may be of the same design as the tool base unit 100 show in U.S. Pat. No. 9,956,677, which is hereby incorporated by reference in its entirety and the tool base unit 100 of the present application may similarly accept the variety of tool heads as discussed in U.S. Pat. No. 9,956,677.

U.S. Pat. Nos. 9,421,682 and 9,776,315 are other examples of multi-headed tool systems. U.S. Pat. Nos. 9,421,682 and 9,776,315 are hereby incorporated by reference in their entireties.

The inflator head portion 200 includes a connection portion 210. The connection portion is configured to connect to the tool base unit and transfer power from the motor of the tool base unit 100 to the tool head portion. The inflator head 200 further includes a fan housing portion 220 which houses a fan (fan 60 in FIGS. 17 and 18 is described further below). The fan draws air in through the inlet 240 and pushes air out through the air outlet 245. In this way, the inflator 10 can be used to inflate and deflate inflatable objects.

FIGS. 3 and 4 show an example of the inflator 10 inflating and deflating a mattress 500. In FIG. 3, the air outlet 245 is engaged with the mattress 500 and the air inlet 240 is directed towards ambient air. In this position, air is drawn in through the air inlet 240 and directed through the air outlet 245 to the mattress 500.

FIG. 4 shows the inflator 10 in a position to deflate the mattress 500. In FIG. 4, the air inlet 240 is engaged with the mattress 500. The air outlet 245 is directed to ambient atmosphere. Accordingly, the inflator 10 draws air in from the mattress 500 through the air inlet 240 and blows it out of the air outlet 245. In this way, it deflates the mattress 500. Although the inflator 10 is shown inflating and deflating a mattress 500 in FIGS. 3 and 4, it is understood that the inflator 10 may similarly be used to inflate or deflate a variety of other inflatable products such as rafts, inter-tubes, beach balls and the like.

FIGS. 5 and 6 illustrate the inflator head 200 alone. FIG. 6 illustrates a portion of the connection section 210 that interfaces with the tool base unit 100. As shown, the connection section 210 includes an intermediate input member 215 which receives a rotary drive from the motor. In particular, the intermediate input member 215 engages with an intermediate output member which is driven by the motor. In this way, rotary drive from the motor in the tool base unit 100 is transferred to the inflator tool head 200.

As shown in FIGS. 1-6, the air inlet 240 has an inlet axis that is generally parallel or concentric with a longitudinal axis of the motor. The air outlet 245 is transverse to the air inlet 240. Additionally, the air inlet 240 is generally transverse to the handle 12.

FIGS. 7-12 illustrate another exemplary embodiment of an inflator 101. The inflator 101 is the same as inflator 10 except with respect to the orientation of the air outlet 345. In particular, the tool base unit 100 is the same as in the inflator 10. Additionally, the inflator tool head 300 is the same as the inflator tool head 200, except that the air outlet 345 of the inflator tool head 300 is disposed to direct air sideways/horizontal as opposed to upwards/vertical.

FIG. 7 illustrates the inflator 101 in the assembled condition with the inflator tool head 300 attached to the tool base unit 100. As shown in FIG. 7, when the tool head 300 is attached to the tool base unit 100, the air outlet 345 is directed to the side. That is, if the tool base unit 100 is placed on a horizontal surface, the air outlet 345 is generally to the side rather than up or down. Put another way, the air outlet 345 is generally transverse to the handle 12. The air outlet 345 is also generally transverse to a longitudinal axis of the motor. The motor produces a rotary drive along the longi-

tudinal axis of the motor. The longitudinal axis of the motor is generally parallel or concentric with a longitudinal axis of the motor housing 13 and is generally transverse to the handle 12.

FIG. 8 illustrates the inflator 101 with the inflator tool head 300 removed. FIG. 9 illustrates the inflator 101 with the air outlet 345 engaged with the mattress 500 in order to inflate the mattress 500. FIG. 10 illustrates the inflator 101 with the air inlet 240 engaged with the mattress 500 in order to deflate the mattress 500. FIGS. 11 and 12 are views of the inflator tool head 300 by itself.

The first embodiment of the inflator 10 in FIGS. 1-6 illustrates the air outlet 245 in a first, upward, direction. The second embodiment of the inflator 101 in FIGS. 7-12 illustrates the air outlet 345 in a second, sideways, direction. The air outlet may also be radially located at various other directions. For example, the air outlet may be directed down instead of up. The air outlet 345 may be directed to the opposite side from that shown in FIGS. 7-12. Additionally, the air outlets 245, 345 may be directed at intermediate angled positions.

FIGS. 13-18 illustrate another exemplary embodiment of an inflator 102. Inflator 102 has the same base unit 100 as the inflators 10 and 101. Inflator 102 has an inflator tool head 400. The inflator tool head 400 is similar to the inflator tool heads 200 and 300. However, the inflator tool head 400 has a rotatable portion such that the air outlet 445 may be rotated 360 degrees to be pointed at various directions. For example, in FIG. 13, the outlet 445 is pointed straight up. In FIG. 14, the outlet 445 has been rotated to be at an angle between horizontal and vertical, roughly 45 degrees from horizontal or within a range of 30 to 60 degrees from vertical or horizontal. FIG. 15 illustrates the outlet 445 pointing to the side/horizontal and FIG. 16 illustrates the outlet pointing horizontally to the opposite side as that shown in FIG. 16. The outlet 445 may be positioned at any angle along about the rotational axis.

FIGS. 17 and 18 illustrate the rotatable structure of the inflator tool head 400. As shown in FIGS. 17 and 18, the inflator tool head 400 has a fan housing portion 420 that is made as a separate part and is rotatable with respect to the connection portion 410. This is in contrast to the inflator tool heads 200 and 300, in which the connection portion 210 and the fan housing portion 220 are made of a single piece.

As shown in FIGS. 17 and 18, the connection portion 410 has a projection 411 and the fan housing portion 420 has a groove 421. The projection 411 engages with the groove 421 to rotatably connect the fan housing portion 420 and the connection portion 410. As also shown in FIGS. 17 and 18, the fan connection portion 410 has an abutting wall 412 and the fan housing portion has an abutting wall 422. The connection portion abutting wall 412 abuts the housing portion abutting wall 422 providing stability to the connection. In some embodiments, these abutting walls 412, 422 provide friction to maintain the relative rotational location of the fan housing portion 420 to the fan connection portion 410. The user can then rotate the fan housing portion 420 by applying pressure to overcome the friction. In this way, the user can change the angular position of the air outlet 445 among the various rotational locations. The amount of friction can be provided by the dimensioning of the connection and fan housing portions 410 and 420; the materials of the connection and fan housing portions 410 and 420 or the abutting walls 412, 422; surface treatments of the connection and fan housing portions 410 and 420 or the abutting walls 412, 422; other methods for increasing or controlling friction or a combination of the same. For

example, the abutting walls 412, 422 may be made to tightly abut one another in order to increase friction. As another example, facing surfaces of the abutting walls 412, 422 may be coated with a high friction material such as rubber in order to increase the friction between the connection housing 410 and fan housing portions 420.

The fan housing portion includes both the air inlet 240 and the air outlet 445. The air inlet 240 is at or on a rotational axis of the fan housing portion 420. Accordingly, although the air inlet 240 of the exemplary embodiment rotates along with the fan housing portion 420, the axis of the air inlet 240 changes none if on the rotational axis or only slightly if slightly off of the rotational axis. Thus, the overall position of the air inlet 240 is relatively the same in all positions of the fan housing portion 420. In other exemplary embodiments, the air inlet 240 may be disposed offset from the rotational axis of the fan housing portion 420 such that the position of the axis of the air inlet 240 also changes when the fan housing portion 420 rotates.

In contrast, the air outlet 445 is at an outer circumference of the fan housing portion 420. Additionally, the air outlet 445 has an axis that is transverse to a rotational axis of the fan housing portion 240. Accordingly, the position of the air outlet 445 changes along with the rotational position of the fan housing portion 240.

The rotational position of the fan housing portion 420 may alternatively or additionally be provided by a biasing assembly. In the exemplary embodiment, there is ball 430 biased by a spring 431 from the connection portion 410 towards the fan housing portion 420. The ball 430 contacts the fan housing portion 420 to maintain the fan housing portion 420 in a rotational location. A user may provide a force to overcome the force provided by the ball 430 to move from the particular rotational location to another rotational location. The fan housing portion 420 may include recesses at particular locations that receive the ball 430. The recesses 435 are located at particular specific locations which may be useful to a user. For example, the recesses 435 may be placed at four locations about the fan housing portion 420, ninety-degrees apart. That would allow the fan housing portion 420 to be specifically set at those four specific and discrete locations, such as up, down, left and right. In other embodiments, there may be eight recesses 435, each forty-five degrees apart. In that instance, four additional locations could be provided between the up, down, left and right positions mentioned above. There may be more or less recesses 435 providing a fewer or greater number of discrete locations for the disposition of the fan housing portion 420 and thus the air outlet 445. There may also be more than one biasing assembly. Additionally, the biasing assembly may be different than a ball biased by a spring. For example, another biasing member may be used instead of a spring and instead of a ball there may be another type of projection or rigid element that engages with the recesses 435. The biasing member and rigid element may also be one piece so that the biasing assembly is a one piece member that serves the same function. For example, this could be a compressible projection.

Accordingly, as shown in FIGS. 12-18, the position of the air outlet 445 is rotatable relative to the handle 12 and is rotatable about a rotation axis. As shown, the rotation axis may be generally concentric or parallel to the longitudinal axis of the motor and generally transverse to the handle 12.

FIGS. 17 and 18 provide cross-sectional views of the inflator head 400 so that internals of the inflator head 400 can be seen. As shown in FIGS. 17 and 18, the inflator head 400 includes a drive connector 70. The drive connector 70

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receives rotational motion from the motor. A spindle **75** is driven through the drive connector **70** and is connected to a fan **60**. Accordingly, the motor drives the fan **60**. Internals of the inflator heads **200** and **300** are the same as that for the inflator head **400** except that the inflator heads **200** and **300** do not include the rotational features of inflator head **400**. Accordingly, inflator heads **200** and **300** similarly include drive connector **70**, spindle **75** and fan **60**. Groove **421**, projection **411** and abutting walls **412**, **422** and the biasing assembly may be removed in inflator heads **200** and **300**. However, one or more of these components may also be maintained in the inflator heads **200**, **300** in order to provide rigidity. For example, the structure of abutting walls **412**, **422** may be included in the inflator heads **200**, **300** in order to provide rigidity for the inflator heads even if they lack a rotation feature. In that case, the abutting walls **412**, **422** may be joined or formed as one integral part.

FIGS. **19** and **20** illustrate inflator attachment holders **500**. The inflator attachment holders **500** are attached to either side of the connection portion **410**. The inflator attachment holders **500** hold inflator attachments **510** which can be slid over the air inlet or air outlet so as to interface with various inflatables. FIG. **19** illustrates the attachment holders **500** holding the inflator attachments **510** and FIG. **20** illustrates the attachment holders **500** when empty.

In the exemplary embodiments, the inflators **10**, **101** and **102** are formed by a tool base unit and a removable inflator tool head unit. The inflators may also be made as a stand-alone tool where the inflator tool head is integrated with a base unit in a fixed manner rather than being separable.

Although the present technology has been described in detail for the purpose of illustration based on what is currently considered to be the most practical and preferred embodiments, it is to be understood that such detail is solely for that purpose and that the technology is not limited to the disclosed embodiments, but, on the contrary, is intended to cover modifications and equivalent arrangements that are within the spirit and scope of the appended claims. For example, it is to be understood that the present technology contemplates that, to the extent possible, one or more features of any embodiment can be combined with one or more features of any other embodiment.

Additionally, while the exemplary embodiment is described with respect to an inflating tool, the methods and configurations may also apply to or encompass other power tools.

What is claimed is:

1. An inflator, comprising:

a tool base unit, the tool base unit comprising a handle, a motor and a trigger; and
an inflator tool head removably attached to the tool base unit;

wherein the inflator tool head comprises an air inlet, an air outlet and a fan configured to draw air in through the air inlet and blow air out through the air outlet;

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wherein the inflator tool head comprises a connection portion and a fan housing portion;
wherein the fan is disposed in the fan housing portion; and
wherein the fan housing portion is rotatable with respect to the connection portion.

2. The inflator according to claim 1, wherein the fan housing portion includes the air outlet.

3. The inflator according to claim 2, wherein the fan housing portion includes the air inlet.

4. The inflator according to claim 2, wherein the fan housing portion is configured to be held in place in a variety of rotational positions with respect to the connection portion.

5. The inflator according to claim 4, wherein the fan housing portion is configured to be held in place by friction.

6. The inflator according to claim 4, wherein the fan housing portion is configured to be held in place by a biasing assembly.

7. The inflator according to claim 6, wherein the fan housing portion is configured to be held in place at plurality of distinct rotational positions.

8. The inflator according to claim 7, wherein the plurality of distinct rotational positions comprises at least four distinct rotational positions.

9. The inflator according to claim 8, wherein the connection portion houses a connector which receives drive from the motor.

10. A high-volume, low pressure inflator, comprising:

a housing;

a handle formed by the housing;

a motor housed in the housing;

a trigger configured to activate the motor;

an inflator head portion;

an inlet having an inlet axis that is generally parallel or concentric with a longitudinal axis of the motor;

an outlet which is transverse to the inlet;

a fan configured to draw air in through the inlet and blow air out through the outlet;

wherein the inlet is generally transverse to the handle;

wherein the position of the outlet is rotatable relative to the handle;

wherein the inflator head portion comprises a connection portion and a fan housing portion;

wherein the fan is disposed in the fan housing portion; and

wherein the fan housing portion includes the outlet and is rotatable with respect to the connection portion.

11. The high-volume, low pressure inflator of claim 10, wherein the outlet is rotatable about a rotation axis;

wherein the rotation axis is generally concentric or parallel to the longitudinal axis of the motor; and

wherein the rotation axis is generally transverse to the handle.

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